

Hsiu-I Liao

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Summary

ICPC Asia Taiwan medalist and NYCU computer science student minoring in applied mathematics, with strong skills in quantitative modeling and statistics. Hands-on experience in quantitative research and trading, with improvements in Sharpe ratio (up to 1.64) and out-of-sample prediction accuracy. Proficient in Python/C++, ML frameworks, and low-latency trading system design.

Education

National Yang Ming Chiao Tung University (NYCU)

M.S. in Network Engineering

Hsinchu, Taiwan

Sep. 2026 (Expected) – Jun. 2028

National Yang Ming Chiao Tung University (NYCU)

B.S. in Computer Science, Minor in Applied Mathematics

Hsinchu, Taiwan

Sep. 2022 – Jun. 2026 (Expected)

- Overall GPA: 3.96/4.3
- Recipient of the Elite Student Award, awarded NT\$100,000. (Fall 2025)
- Coursework: Probability, Statistics, Stochastic Processes, Optimization, Machine Learning, Reinforcement Learning, Algorithms, Advanced Calculus, Investments.

Work Experience

A&R Research Capital

Xinyi District, Taipei, Taiwan

Quantitative Research Intern

Jun. 2025 – Aug. 2025

- Designed a short-only momentum strategy by detecting algorithmic order bursts in medium-sized ETF constituent prints; a 6-month backtest (Jun.–Dec. 2024, including 14.25 bps per-side commissions) achieved a **Sharpe ratio of 1.64**, a **58% win rate**, and a **maximum drawdown of 2.7%**.
- Reverse-engineered an ETF index methodology from the prospectus and exchange audits, achieving **40/40 correct constituent predictions** from 2023 to 2025 and reducing weight **RMSE to 0.29%**.
- Developed a cross-asset hedging framework using TAIEX futures and rolling-window CAPM for four trading strategies, improving average **Sharpe ratio by 0.11** under a beta-neutral constraint.
- Benchmarked Elastic Net and linear regression models with RobustScaler, PCA, and SHAP, improving out-of-sample **R-squared by 1.48%**.

Kronos Research

Hsinchu, Taiwan

Quantitative Trading Program Trainee

Sep. 2024 – Jan. 2025

- Developed an asynchronous cryptocurrency trading framework with **Python asyncio** and **aioredis** for low-latency execution.
- Implemented **Publisher/Subscriber** architecture via **Redis pub/sub** for real-time market data distribution.
- Integrated **WooX REST** and **WebSocket** APIs for market and account data streaming.
- Designed a modular **Strategy Executor** supporting CTA and custom strategies.
- Implemented API authentication with **HMAC-SHA256** signing and timestamp validation.

Institute of Information Science at Academia Sinica

Taipei, Taiwan

Summer Internship

Jul. 2023 – Aug. 2023

- Proved an $\Omega(n^2)$ space lower bound for transitivity testing using reduction and dynamic programming, and showed the bound is tight.

Honors & Awards

Academic Achievement Award (Top 5%), Computer Science, NYCU (Spring 2024)

Hsinchu, Taiwan

Gold Medal, WorldQuant Challenge Gold Certificate (2025)

Online

Silver Medal, ICPC Asia Jakarta Onsite Regional Contest (2025)

Jakarta, Indonesia

Silver Medal, ICPC Asia Taichung / Taoyuan Regional Programming Contest (2025, 2022)

Taiwan

Gold Medal, ICPC Asia Taiwan Online Programming Contest (2025)

Online

Silver Medal, ICPC Asia Taiwan Online Programming Contest (2024, 2023, 2022)

Online

7th, National Collegiate Programming Contest (2025)

Taipei, Taiwan

Skills

- Quantitative research, statistical testing, time-series modeling
- Python, C++, asyncio, Redis, REST/WebSocket APIs, algorithmic trading systems
- Languages: Mandarin Chinese (native), English (TOEFL iBT: 90/120)